

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 3 SET B**SUMMER SEMESTER, 2024-2025****DURATION: 20 MINUTES****FULL MARKS: 15**

SWE 4805: Software Verification and Validation

Write all the answers on the question paper.

Answer **all 1 (one)** questions. Figures in the right margin indicate full marks of questions.

ID:

Consider the following Java method that determines whether a student has passed a course.

```
1 public class PassCheck {  
2  
3     public static boolean hasPassed(int marks, int attendance) {  
4  
5         if (attendance < 75)  
6             return false;  
7  
8         if (marks >= 50)  
9             return true;  
10  
11         return false;  
12     }  
13 }
```

(CO1)
(PO1)

The following test cases are available:

Test Case	Marks	Attendance (%)	Expected Output
T1	65	80	true
T2	45	80	false
T3	70	70	false
T4	50	75	true

1. Generate any **six** valid mutants of the given program using the mutation categories covered in class: **Value Mutation, Decision Mutation, and Statement Mutation**. For each mutant, clearly indicate the mutation category and show the modified line of code. 6
2. For the mutants created in part (a), construct a mutation matrix indicating whether each test case **Kills (K)** or **Survives (S)** each mutant. 5
3. Using the mutation matrix obtained in part (b), calculate the **Mutation Score** of the test suite. 4
Clearly show:
 - Number of mutants generated
 - Number of killed mutants
 - Mutation Score formula
 - Final Mutation Score (as a percentage)